Not only Look, but also Listen: Learning Multimodal Violence Detection under Weak Supervision

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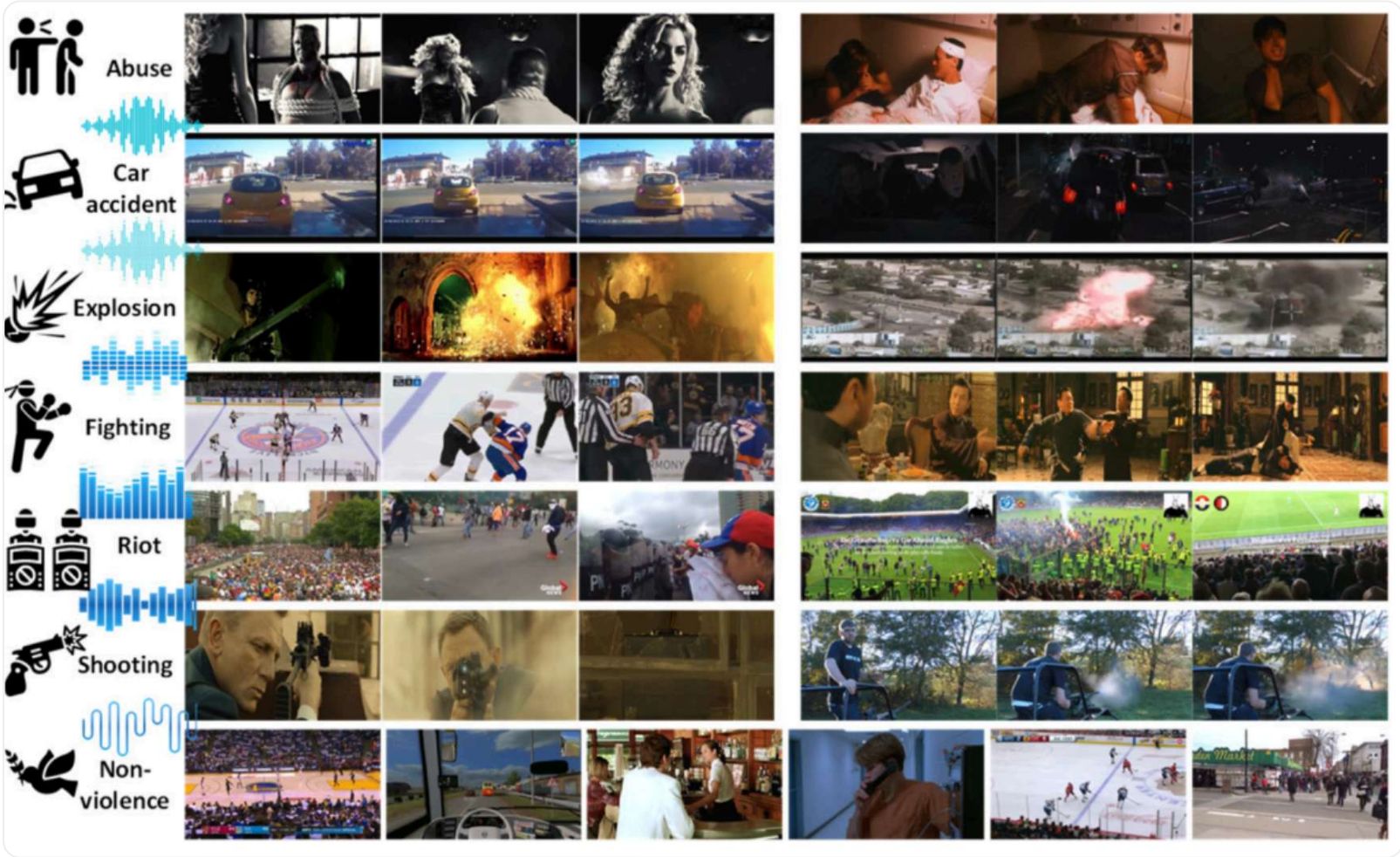


Sample videos from the XD-Violence dataset.

Abstract

Violence detection has been studied in computer vision for years. However, previous work are either superficial, e.g., classification of short-clips, and the single scenario, or undersupplied, e.g., the single modality, and hand-crafted features based multimodality. To address this problem, in this work we first release a large-scale and multi-scene dataset named XD-Violence with a total duration of 217 hours, containing 4754 untrimmed videos

with audio signals and weak labels. Then we propose a neural network containing three parallel branches to capture different relations among video snippets and integrate features, where holistic branch captures long-range dependencies using similarity prior, localized branch captures local positional relation using proximity prior, and score branch dynamically captures the closeness of predicted score. Besides, our method also includes an approximator to meet the needs of online detection. Our method outperforms other state-of-the-art methods on our released dataset and other existing benchmark. Moreover, extensive experimental results also show the positive effect of multimodal input and modeling relationships.

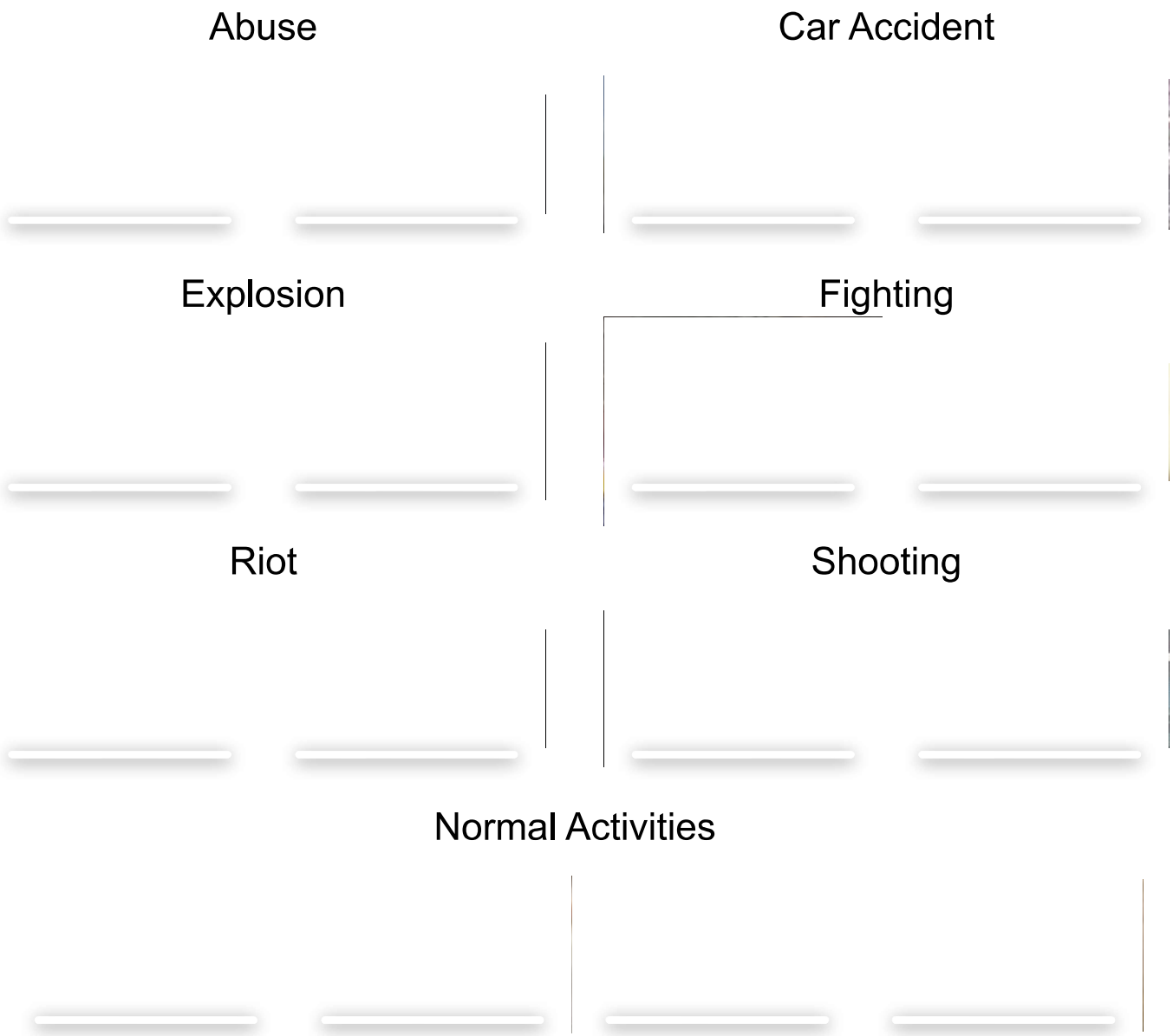
Representation Video

0:00 / 6:18

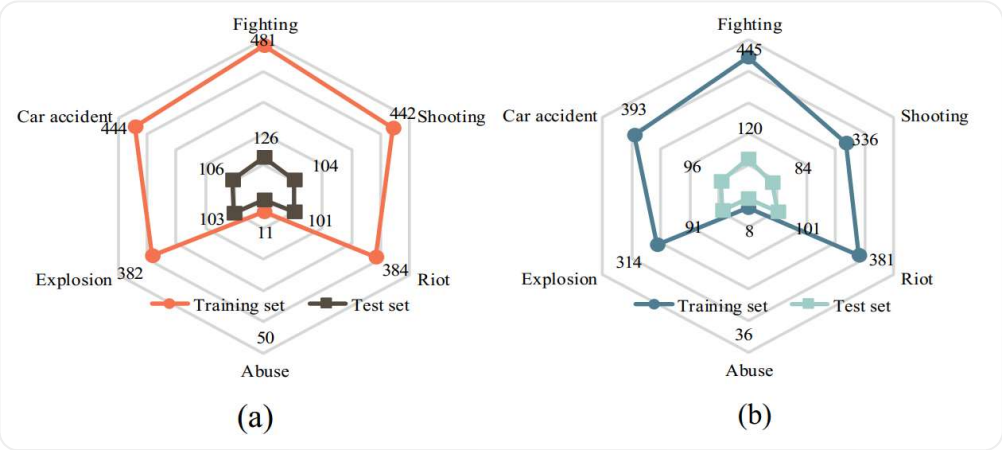
Presentaion

Violence examples

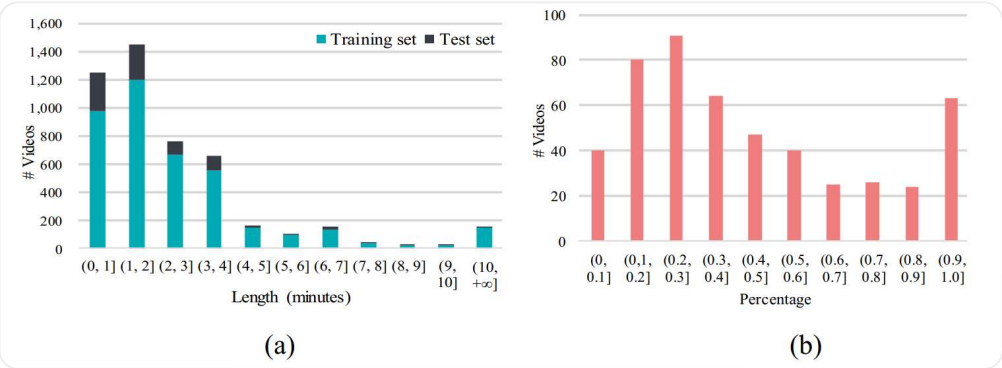
Abuse, Car Accident, Explosion, Fighting, Riot, and Shooting



Dataset Statistics



Dataset Statistics. (a) Distribution of the number of videos belonging to each category according to multi-label. (b) Distribution of the number of videos belonging to each category according to the first label.



Dataset Statistics. (a) Distribution of videos according to length (minutes). (b) Distribution of violent videos according to percentage of violence (in each video) in test set.

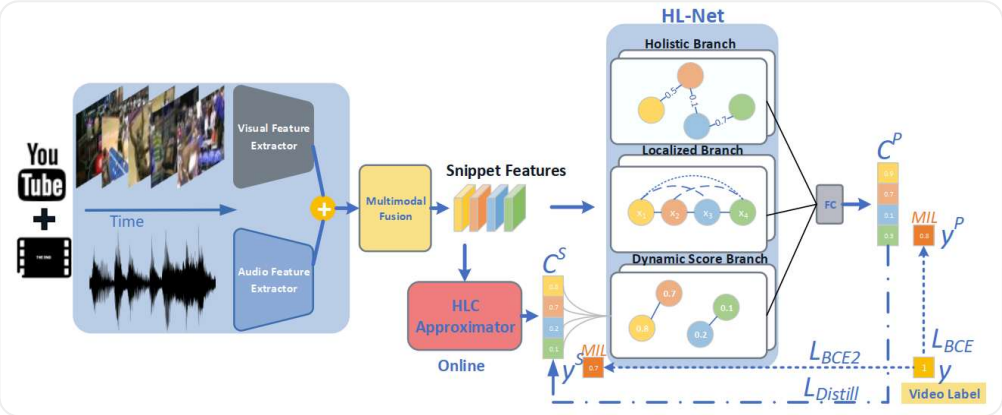
Dataset	#Videos	Length	Source of scenarios	#Violence types	Audio
Hockey [24]	1000	27 min	Ice hockey	1	No
Movie [24]	200	6 min	Movies and sports	1	No
Violent-Flows [15]	246	15 min	Streets, school, and sports	1	Yes*
CCTV-Fights [29]	1000	18 hours	CCTV and mobile cameras	1	Yes*
VSD [4]	25	35 hours	Movies	8	Yes
UCF-Crime [35]	1900	128 hours	CCTV camera	9	No
XD-Violence (Ours)	4754	217 hours	Movies, sports, games, hand-held cameras, CCTV, car cameras, etc.	6	Yes

Comparisons of different violence datasets. ∗ means quite a few videos are silent or only contain background music.

Traits

1) large scale, which is beneficial for training generalizable methods for violence detection;
2) diversity of scenarios, so that violence detection methods actively respond to complicated and diverse environments and are more robust;
3) containing audio signals, making algorithms leverage multimodal information and more confidence;
4) multi labels, We assign multi violent labels (1 ≤ #labels ≤ 3) to each violent video owing to the co-occurrence of violent events. The order of labels of each video is based on the importance of different violent events occurring in the video.

Methodology



The pipeline of our proposed method.

Code

Github

Download

V1.0 Videos

Baidu Netdisk

~~Training Videos~~ ~~[keyword:1ltx]~~ Disabled

Test Videos [keyword:exye]
Test Annotations
ReadMe

AliyunDrive

Training Videos

OneDrive

Trainging Videos_0001-1004
Trainging Videos_1005-2004
Trainging Videos_2005-2804
Trainging Videos_2805-3319
Trainging Videos_3320-3954
Test Videos
Test Annotations

V1.0 Features

Baidu Netdisk

audio features (VGGish) [keyword:i9h2]
visual features (I3D RGB&Flow) [keyword:ou1n]
ReadMe

OneDrive

audio features (VGGish)
visual features (I3D RGB&Flow)

V1.0 PRC Data

precision and recall values of PRC

Paper

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(Paper)

(Supplementary materials)

[arXiv]

Cite

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Contact

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